

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 32

Title: Liver function tests

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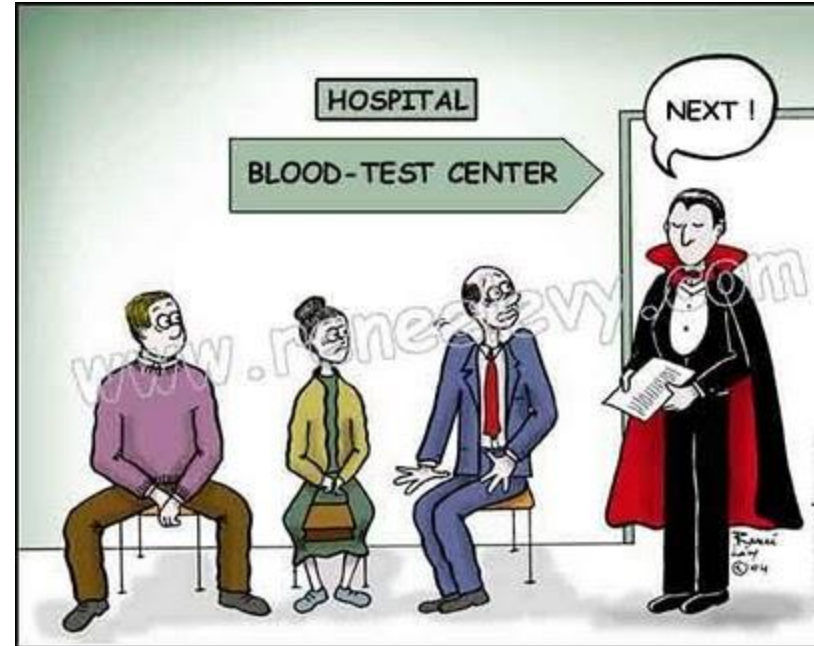
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Liver function tests



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Objectives: Liver function tests

SOM. MK.II.BPM2.3.DM.2.BCHM.1297	List important functions of the liver: Excretion of bilirubin, Synthesis of plasma proteins, Detoxification of ammonia
SOM. MK.II.BPM2.3.DM.2.BCHM.1298	Interpret the following laboratory tests in patients with acute hepatitis, alcoholic liver disease and cholestasis: Serum (total, conjugated and unconjugated) and urine bilirubin, Serum enzymes (ALT, AST, ALP, GGT), Serum proteins (albumin and globulin), Prothrombin time (INR), Serum ammonia, alpha fetoprotein (AFP), alpha1- antitrypsin, ceruloplasmin, serum iron, transferrin and ferritin
SOM. MK.II.BPM2.3.DM.2.BCHM.1299	Compare and contrast changes in liver function tests in hepatocellular diseases and disorders associated with cholestasis
SOM. MK.II.BPM2.3.DM.2.BCHM.1300	Differentiate between acute and chronic liver disease based on liver function tests
SOM. MK.II.BPM2.3.DM.2.BCHM.1301	Explain molecular mechanisms for the occurrence of edema, icterus, ascites, hepatic encephalopathy, and bleeding tendency in patients with liver disease.

Recommended Reading

- **Evaluating the Patient With Abnormal Liver Tests**
http://www.medscape.com/viewarticle/710045_2
- Harrison's Principles of Internal Medicine: Chapter 348: Evaluation of Liver Function; (Available on Access Medicine) [Evaluation of Liver Function | Harrison's Principles of Internal Medicine, 22nd Edition | AccessMedicine | McGraw Hill Medical](#)

Functions of liver

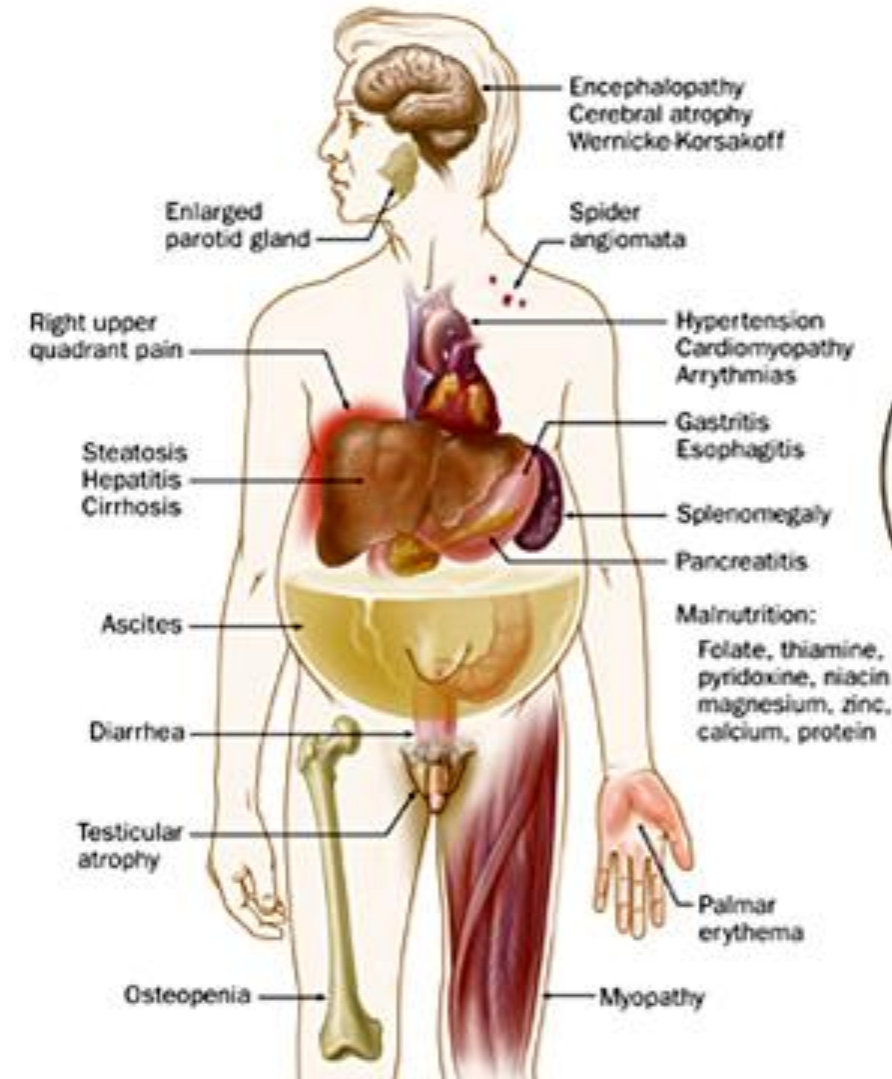
- Excretion, plasma protein synthesis, blood clotting, metabolism and detoxication functions
- Clinical assessment by a panel of lab tests
- Panel of lab tests: Liver function tests (LFTs); Liver chemistry tests; Liver tests
- LFTs indicate extent of anatomical and physiological damage
- No single test capable of indicating functional integrity of liver

History

- High risk behavior
 - IV drug abuse, alcohol abuse, high risk sexual activity
- Systemic illness
 - Diabetes, obesity, cancer
- Medications
 - Acetaminophen, statins
- Family history of liver disease
 - Hemochromatosis, Wilson disease, α_1 -antitrypsin deficiency

Physical exam

- Jaundice
 - Acute hepatitis, biliary obstruction or advanced chronic liver disease
- Abdominal pain and fever
 - Cholecystitis, or inflammatory liver disease
- Stigmata of chronic cirrhosis
 - Spider angioma, palmar erythema, gynecomastia, testicular atrophy
 - Imbalance in testosterone: **estrogen**
- Complications of liver disease
 - Advanced liver disease
 - **Encephalopathy, ascites, GI bleeding (esophageal varices), coagulation disorders**



Spider angioma



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Eyes
• Jaundice

Chest
• Gynaecomastia (in men)
• Breast atrophy (in women)



C Leukonychia

Hands
• Leukonychia (white nails) (C)
• Palmar erythema (B)
• Clubbing



B Palmar erythema



A Spider naevi

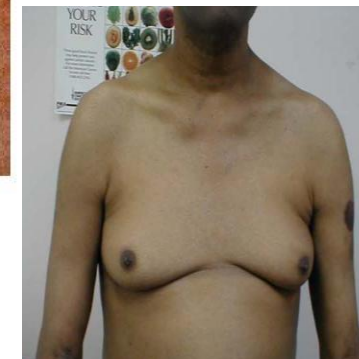
Upper half of body (above umbilicus)
• Spider naevi (A)

Abdomen
• Splenomegaly
• Hepatomegaly (but liver may be small)
• Dilated collateral vessels around umbilicus

Genitalia
• Testicular atrophy

Legs
• Oedema
• Hair loss

General
• Skin pigmentation
• Loss of body hair
• Bruising



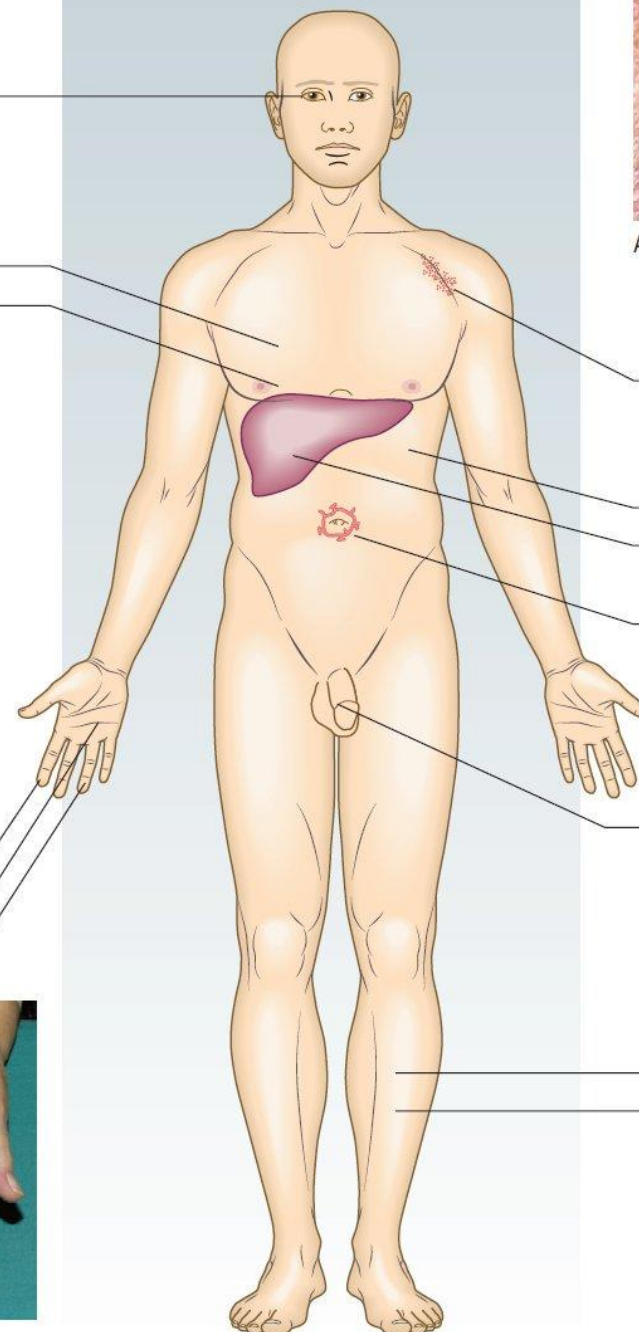
Gynecomastia

Portal hypertension



Caput medusae

11



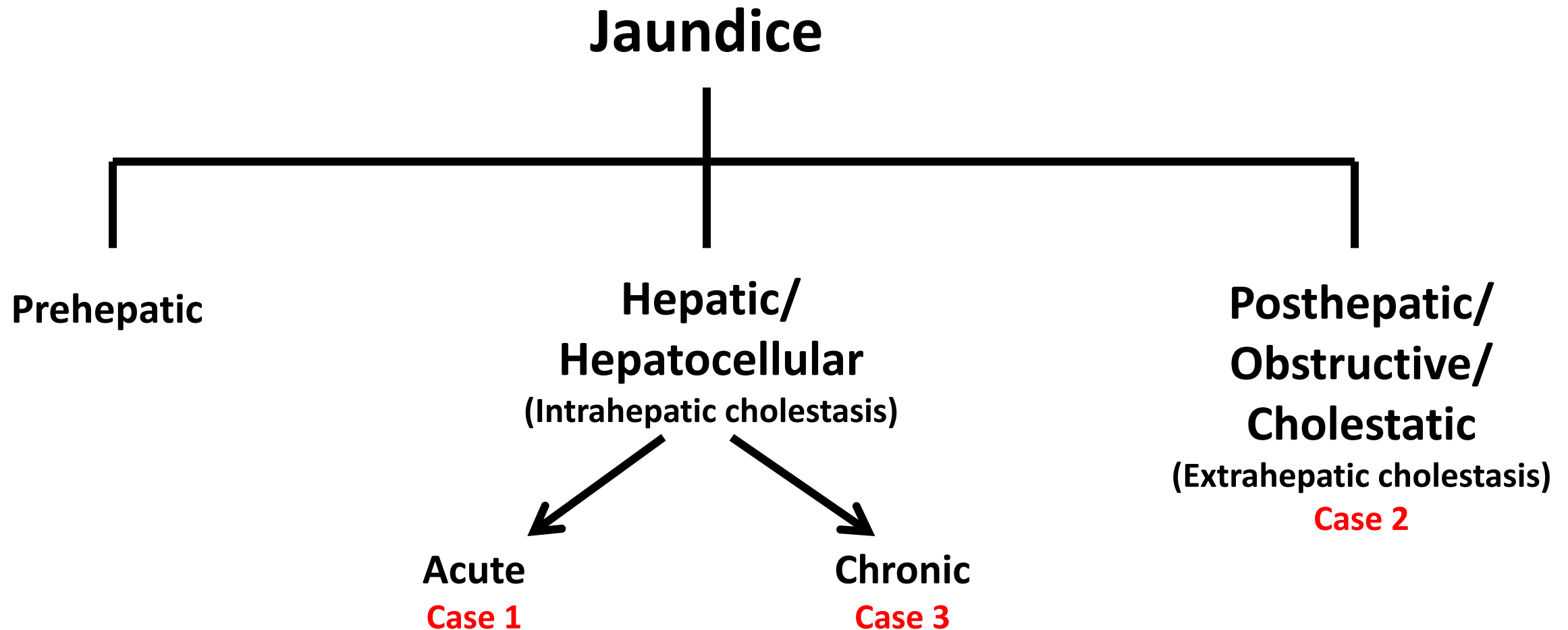
Features of chronic liver disease.

Source : Macleods Clinical Examination 13th Ed (2013)

Jaundice, ascites

SOM. MK.II.BPM2.3.DM.2.BCHM.1301

Liver function tests in patients with jaundice

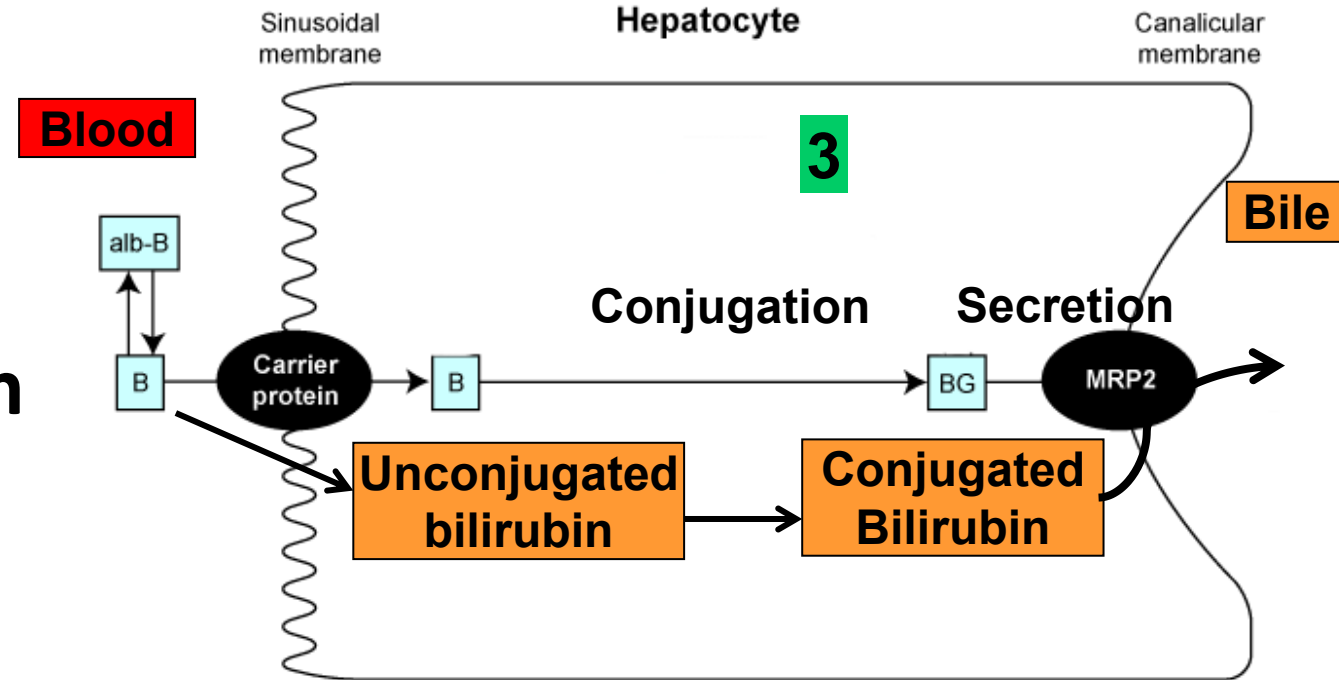


Liver function tests (LFTs)

- Tests for excretory function
 - Serum bilirubin fractions (total, conjugated and unconjugated)
- Serum Enzyme levels
 - Enzymes that indicate hepatocellular damage (ALT and AST)
 - Enzymes that indicate cholestasis (ALP and GGT)
 - Gamma glutamyl transferase (GGT)
- Tests for synthetic function
 - Serum albumin
 - Clotting factors (prothrombin time)
 - Serum alpha-fetoprotein (AFP)
- Tests for metabolic functions
 - Serum ammonia (Hepatic encephalopathy)
 - Plasma glucose
 - Lipid metabolism

Bilirubin metabolism in liver

- Uptake of Unconjugated bilirubin by liver
- **Conjugated bilirubin formed in hepatocytes**
- **Conjugated bilirubin secreted into biliary canaliculi**



Serum bilirubin: Hepatic jaundice

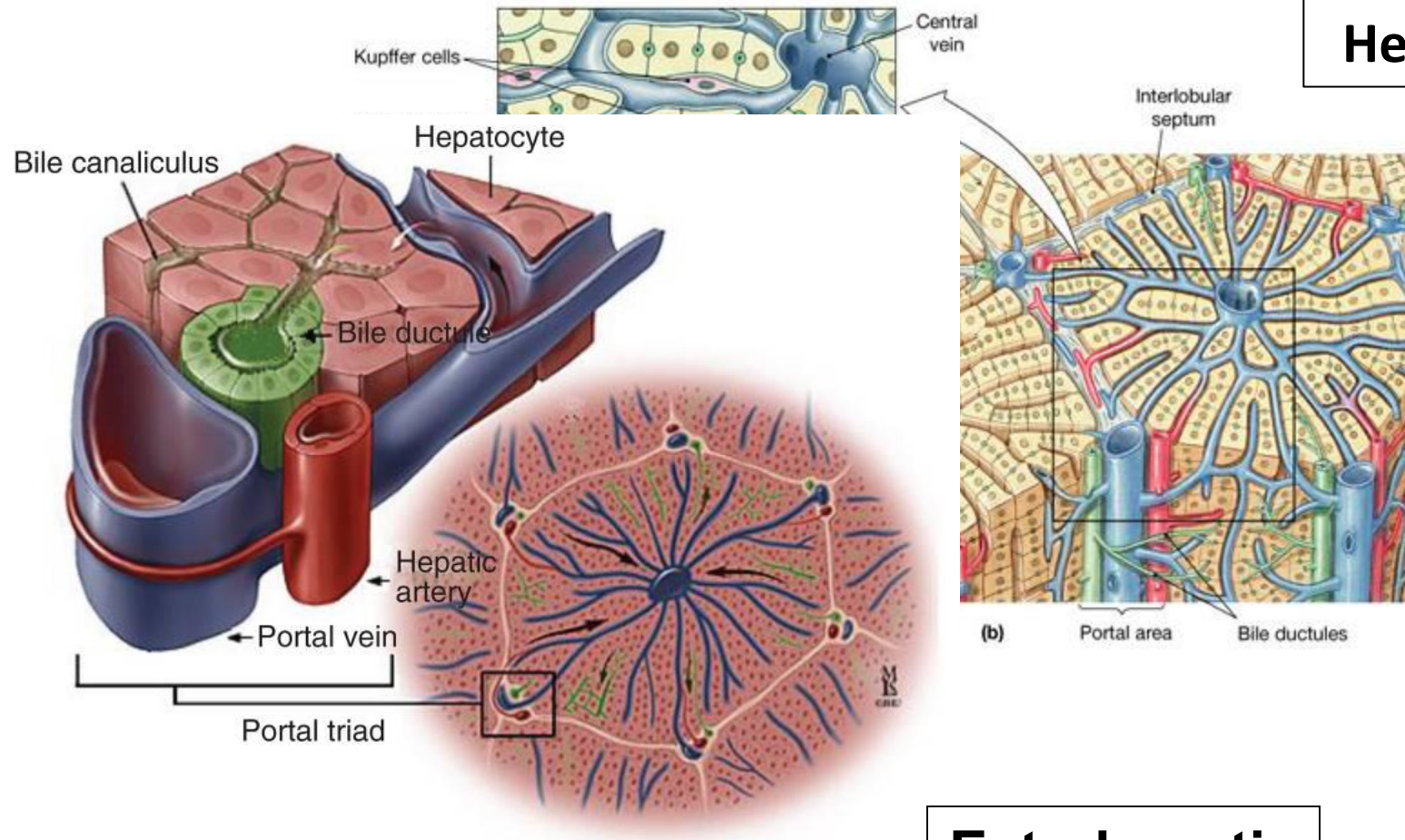


- Conjugated bilirubin in hepatocyte → Biliary secretion → intestine
- **Hepatic jaundice (acute and chronic liver disease):**
 - Elevated serum total, conjugated and unconjugated bilirubin (**mixed**)
 - Elevated Serum Unconjugated bilirubin: Impaired uptake and conjugation
 - Elevated Serum Conjugated bilirubin: Inflammation causes **intrahepatic cholestasis** → **Impaired secretion of conjugated bilirubin into bile**
 - **Conjugated bilirubin in urine (orange-yellow/ dark urine)**

Hepatic/Hepatocellular jaundice

**Intrahepatic
cholestasis**

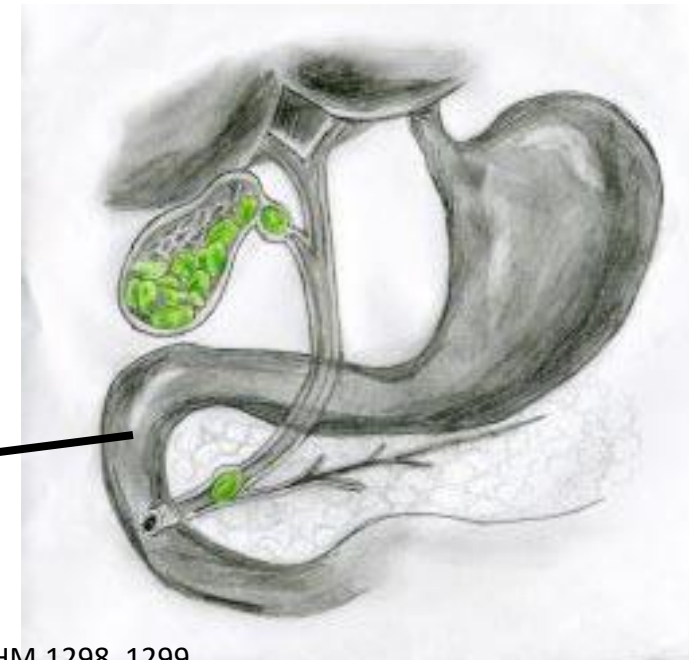
**Hepatocyte inflammation
compresses bile duct/
Chronic cirrhosis**



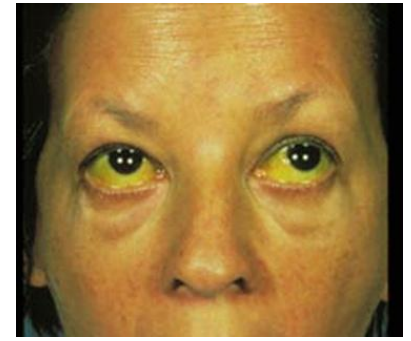
**Posthepatic/Obstructive
/Cholestatic jaundice**

**Extrahepatic
cholestasis**

**Stones in common
bile duct/ Cancer
head of pancreas**



Serum bilirubin: Posthepatic/Obstructive/ Cholestatic jaundice

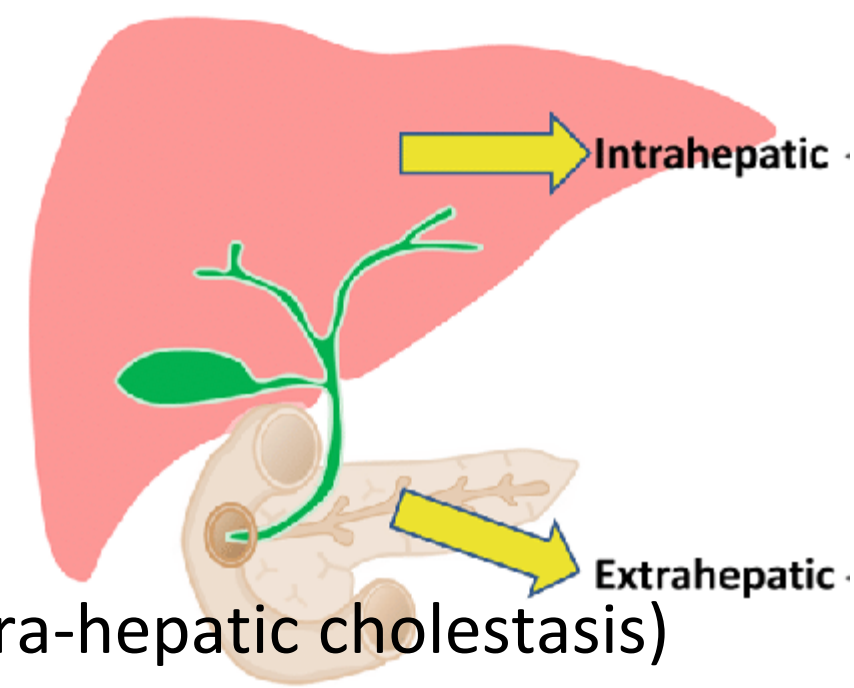


- Cholestatic disease (extrahepatic cholestasis)
 - Elevated Serum **conjugated bilirubin**
 - **Conjugated bilirubin** not secreted into bile
 - **Conjugated bilirubin** in urine (**orange-yellow/ dark urine**)
 - Fecal Stercobilin absent (clay/pale feces)
 - Urine Urobilinogen: Absent
 - Prolonged cholestasis, mild elevation of serum unconjugated bilirubin (liver cell damage)



Serum bilirubin

- Hepatic jaundice (Intra-hepatic cholestasis)
 - **Elevated serum conjugated and unconjugated (Mixed)**
 - Acute: Viral hepatitis and drug toxicity and
 - Chronic: Chronic cirrhosis
- Posthepatic/ Bile duct obstruction/Cholestatic (Extra-hepatic cholestasis)
 - **Serum Conjugated bilirubin>>Unconjugated**
 - Stone in common bile duct
 - Tumors around common bile duct or head of pancreas
- **Isolated hyperbilirubinemia (Conjugated or unconjugated):**
 - **Normal enzyme levels (ALT/AST and ALP)**
 - **Gilbert, Crigler-Najjar, and Dubin-Johnson syndromes (Inherited disorders of bilirubin)**
- Prehepatic/hemolytic jaundice
 - Serum Unconjugated bilirubin>>> Conjugated bilirubin
 - Normal ALT/AST and ALP



Enzymes in serum

- Enzyme levels in serum differentiate between
 - Hepatocellular damage/ Hepatic jaundice
 - Posthepatic/ Obstructive/ Cholestatic jaundice
- Degree of rise indicates extent and severity of damage

Enzymes indicating hepatocellular injury (Hepatic jaundice)

- **Aminotransferases** (Transaminases): Use PLP as coenzyme
 - ALT (alanine aminotransferase): Cytosolic
 - AST (aspartate aminotransferase): Mitochondrial
- Liver amino acid metabolism
- Elevated in Liver diseases
 - Acute: Viral hepatitis, drug induced hepatitis, and
 - Chronic: Long-standing cirrhosis
- Acute liver disease: **ALT levels** >>>AST levels
- Long-standing chronic alcoholic cirrhosis: **AST levels** >>ALT levels (AST: ALT ratio 2:1 or higher)
- Note: Enzyme levels normal in Gilbert syndrome and Dubin Johnson syndrome

Enzymes indicating cholestasis

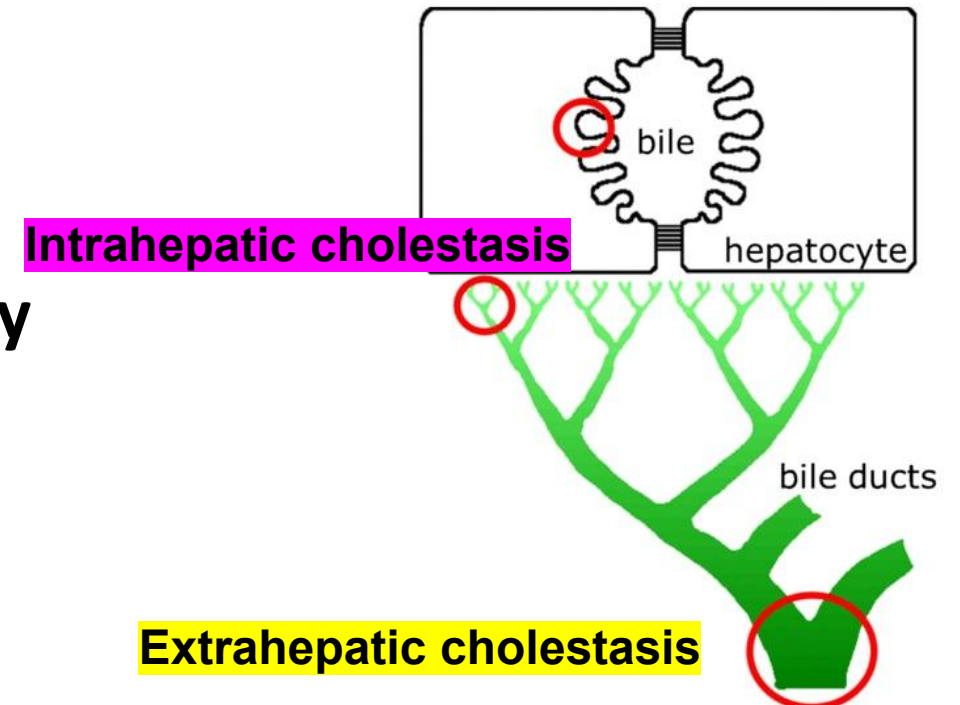
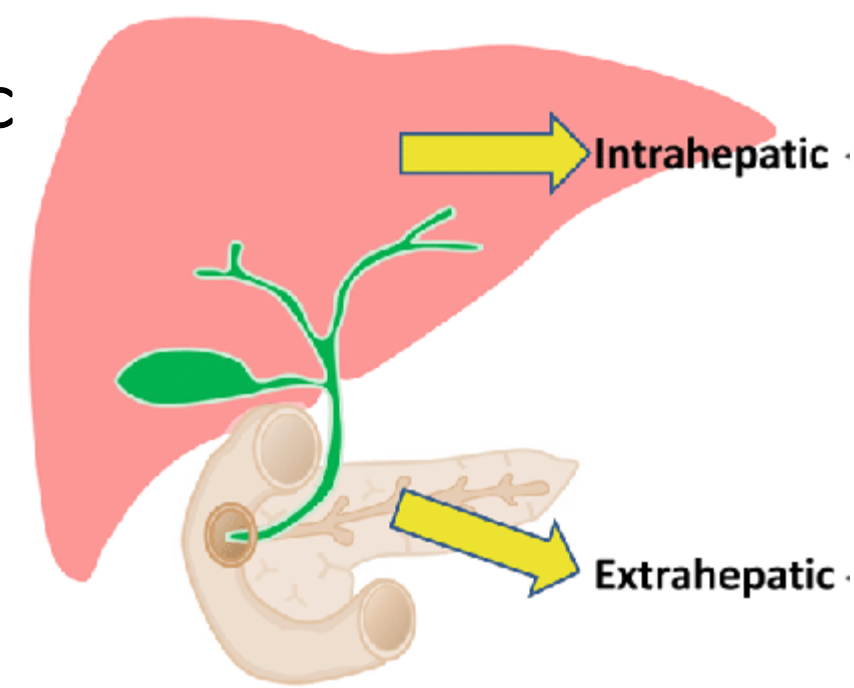
- Cholestasis (intra-hepatic and extra-hepatic)
 - Alkaline phosphatase (ALP)
 - Gamma glutamyl transferase (GGT)
- Alkaline phosphatase (ALP)
 - Secreted by biliary canaliculi
 - **Highly Elevated in extra-hepatic cholestasis (Biliary disease/ Posthepatic)**
 - Mild elevation in intra-hepatic cholestasis (liver disease/ Hepatic)
 - Also elevated in pregnancy, normal growing children and bone diseases

Enzymes indicating cholestasis

- Gamma glutamyl transferase (GGT)
 - Secreted by biliary ducts
 - Induced by alcohol
 - Marker of alcohol consumption (Isolated increase in GGT)
 - Differentiate hepatic and non-hepatic causes of raised ALP
- ALP and GGT elevated in cholestasis
 - Elevated ALP with jaundice (Signs of liver disease/cholestasis)
 - Extra-hepatic cholestasis – very high levels
 - Intra-hepatic cholestasis (Mild elevation); Aminotransferase levels elevated
- **Elevated ALP** with normal GGT: indicates non-biliary ALP (bone, placenta)
- Isolated **elevation in GGT** and normal ALP: Alcohol use or drugs that induce GGT synthesis

Differentiate Obstructive jaundice and Hepatic jaundice

- Abdominal ultrasound assess biliary tree
 - Dilation of biliary tree = **Extrahepatic cholestasis**/ Posthepatic/ **Obstructive**/ Cholestatic jaundice (**Very high ALP and GGT**; Mild elevation of ALT and AST)
 - No biliary tree dilation= **Intra-hepatic cholestasis**/ **Hepatic**/ hepatocellular jaundice (Slight increase ALP and GGT; **Very High ALT and AST**)



Tests for synthetic function of liver

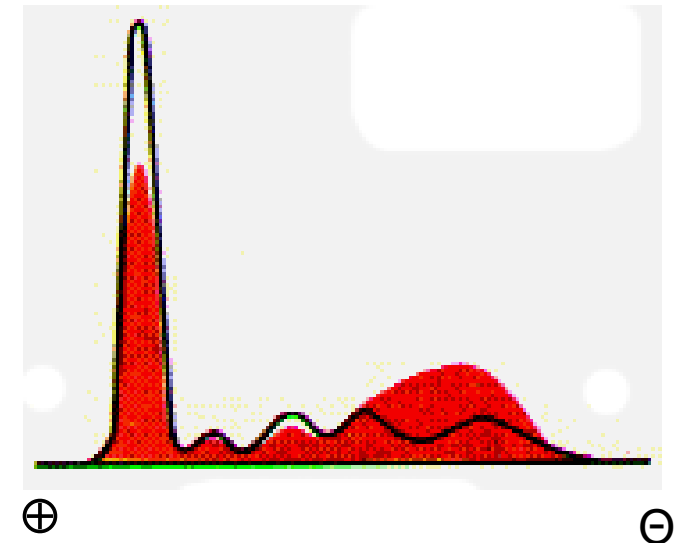
- All plasma proteins, except γ -globulins synthesized by liver
- Serum albumin
 - Only Liver synthesizes albumin
 - Maintains capillary colloidal osmotic pressure
 - Long-standing **(chronic)** liver disease: **Low** serum albumin
 - Responsible for **ascites** and **pedal edema** in long-standing liver disease
 - Affects binding of hormones, calcium and drugs (remember, albumin is major transport protein)



Ascites

Serum albumin and globulin

- In acute hepatitis, normal serum albumin levels
- **Decrease in serum albumin** indicates long-standing (chronic) liver disease, in conjunction with other LFTs
- Serum γ -globulins are elevated in cirrhosis (IgA) resulting in **β - γ bridging** in serum protein electrophoresis



Clotting factors (prothrombin time)

- Liver synthesizes most clotting factors
- Post-translational gamma-carboxylation of glutamic acid of vitamin K dependent clotting factors (II, VII, IX, X)
- **Increased prothrombin time**
- Coagulopathy in liver disease (increased bleeding tendency)
 - Reduced clotting factors synthesis (hepatocellular damage)
 - Impaired modification of vitamin-K dependent clotting factors

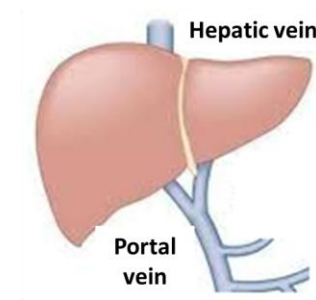
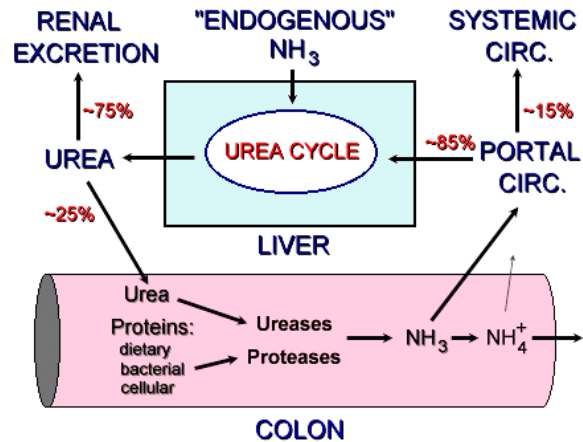
Clotting factors (prothrombin time)

- Coagulopathy in obstructive jaundice/ bile duct obstruction
 - Reduced bile salt entry into intestine impairs fat-soluble vitamin absorption (vitamin K)
 - Impairs post-translational γ -carboxylation of vitamin K dependent clotting factors (II, **VII**, IX and X)
- Monitor prothrombin time (INR) if liver biopsy or surgery planned

Tests for metabolic function: Carbohydrate metabolism

- Plasma glucose
 - Liver functions as 'glucostat'
 - Liver stores glucose as glycogen when blood glucose level is high and releases glucose by glycogenolysis, when blood glucose level is low (in between meals)
 - Chronic liver disease can alter plasma glucose level
 - **Hypoglycemia** or hyperglycemia

Tests for metabolic function: Ammonia

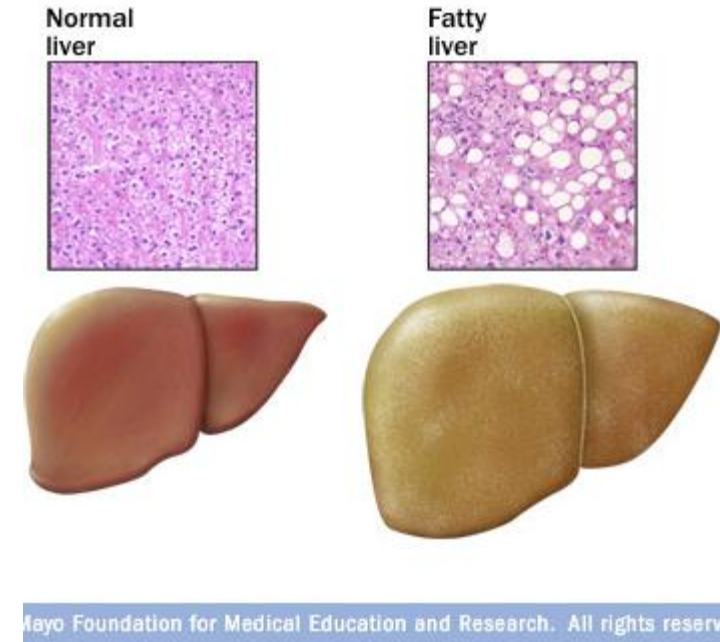


Cirrhosis increases the amount of ammonia entering systemic circulation

- Liver detoxifies Ammonia to urea
- Fulminant (Advanced) end-stage liver disease, impaired urea formation, blood ammonia levels rise
- Results in **hepatic encephalopathy** and **altered consciousness (Coma); Asterixis (flapping tremor)**
- Causes: Neurotransmitter imbalance and TCA cycle depleted of α -ketoglutarate
- GI Bleeding (hematemesis) precipitates **encephalopathy**

Tests for metabolic function: Lipid metabolism

- Liver secretes VLDL and transports TAG to peripheral tissues
- ‘Fatty liver’: Excessive **TAG** (triglyceride) deposits in hepatocytes
- Contributory factors in alcoholism
 - Decreased β -oxidation due to $\uparrow$ NADH/NAD⁺ ratio
 - Increased fatty acid and TAG synthesis
 - Decreased VLDL secretion (hepatotoxic action of ethanol)
- Fatty liver is also seen in obesity, diabetes mellitus

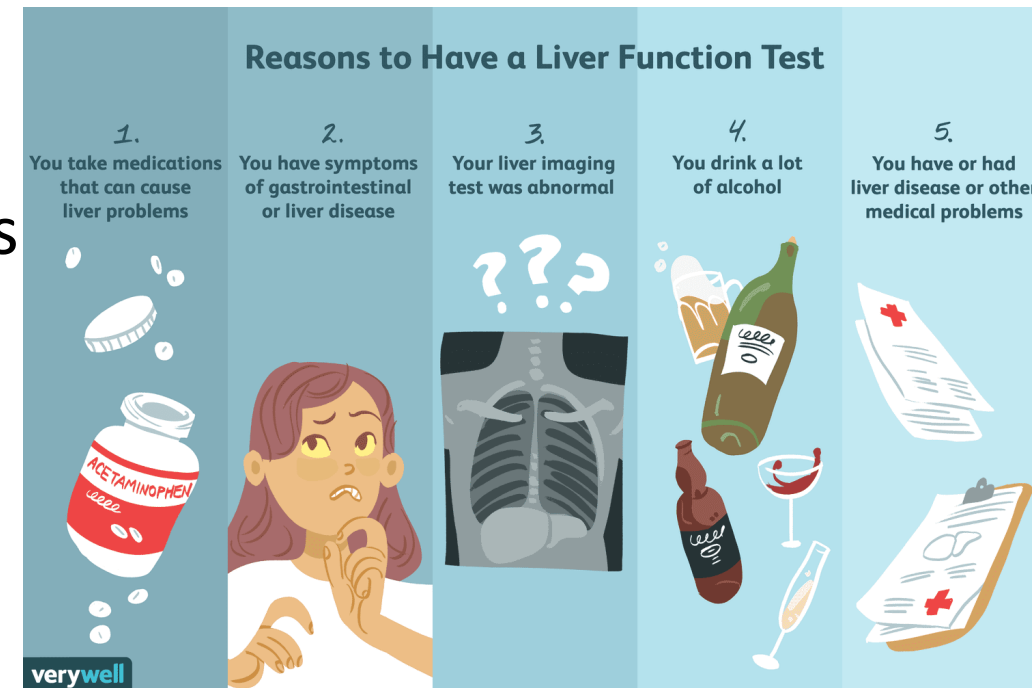
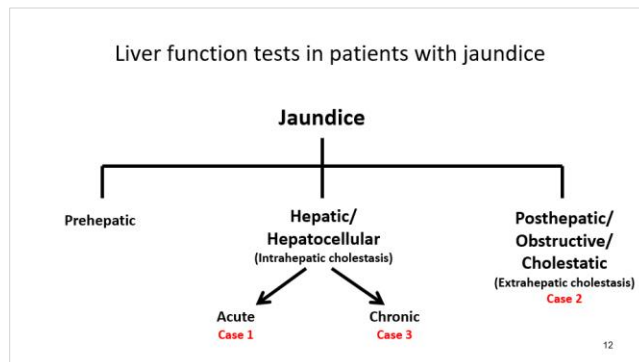


Special tests for evaluation of liver function

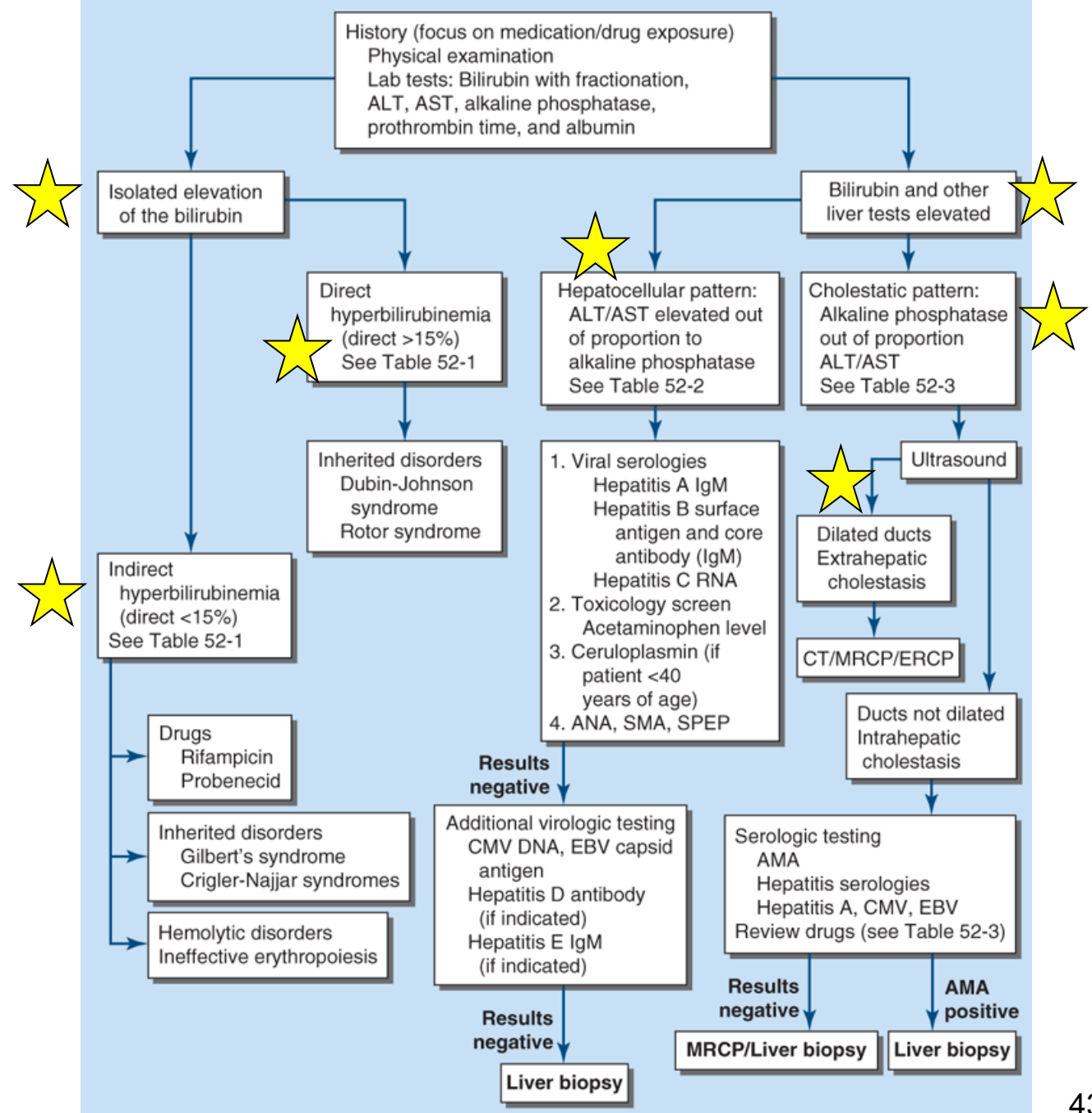
- Serum alpha fetoprotein (AFP): Tumor marker for liver cancer
- Serum iron, transferrin and ferritin
 - Hemochromatosis (Iron overload)
- Serum ceruloplasmin
 - Wilson disease (Copper overload)
- Serum α_1 -antitrypsin
 - Inherited α_1 -antitrypsin deficiency (Lung and liver)

When do we use LFTs?

- Assess intensity and type of jaundice (hepatic or obstructive)
- Assess extent of damage to hepatocytes by viruses, malignancy, drugs
- Differentiate between acute and chronic liver disorders (Serum albumin/ enzymes)
- Assess metabolic functions of liver
- Monitor drug therapy of hepatotoxic drugs
- Follow up and prognosis of liver diseases



Approach to a patient with jaundice



Case report 1

A 40-year-old man has a 3-day history of fatigue and anorexia.
On examination shows scleral icterus (jaundice).

Laboratory parameter	Patient values	Reference range
Serum total bilirubin	5 mg/dL	0.1-1.0 mg/dL
Serum conjugated (direct) bilirubin	3 mg/dL	0.0-0.3 mg/dL
Urine color; Urine bilirubin	Orange yellow; Bilirubin: Present	
ALT	600	10-40 U/L
AST	450	12-38 U/L
ALP	140	25-100 U/L
GGT	100	5-40 U/L
Serum albumin	5 g/dL	3.5-5.5 g/dL
Prothrombin time	25 seconds	11-15 seconds

Case report 1

- A 40-year-old man has a 3-day history of fatigue and anorexia
- On examination, scleral icterus (jaundice)
 - Serum total bilirubin = 5mg/dL
 - Serum conjugated bilirubin = 3mg/dL (Mixed)
 - Urine: Orange yellow (positive for bilirubin)
- ALT >> AST- 600/450 ↑ ↑ ↑ ↑ (Massive hepatocellular damage)
- ALP and GGT- 140/100 ↑ (Intrahepatic cholestasis due to inflammation; confirm by ultrasound)
- Serum albumin – normal (indicates acute liver damage)
- Prothrombin time - ↑ (decreased capacity of liver to synthesize clotting factors)
- **Acute** hepatocellular disease/ **Acute hepatic jaundice**
- Causes: Acute Hepatitis – viral or drug (acetaminophen ...) induced

Case report 2

A 50-year-old woman has 2-month history of worsening jaundice. Urine tea-colored for the past month. Lost 30 lb in past year despite good appetite.

Laboratory parameter	Patient values	Reference range
Serum total bilirubin	6 mg/dL	0.1-1.0 mg/dL
Serum conjugated (direct) bilirubin	5.5 mg/dL	0.0-0.3 mg/dL
Urine color; Urine bilirubin	Orange yellow; Bilirubin: Present	
ALT	100	10-40 U/L
AST	80	12-38 U/L
ALP	540	25-100 U/L
GGT	300	5-40 U/L
Serum albumin	5 g/dL	3.5-5.5 g/dL
Prothrombin time	25 seconds	11-15 seconds

Case report 2

- A 50-year-old woman has 2-month history of worsening jaundice
- Urine tea-colored for the past month.
- Lost 30 lbs in past year despite good appetite.
- Laboratory tests
 - Serum total bilirubin 6mg/dL
 - Serum conjugated bilirubin 5.5mg/dL
 - Urine bilirubin positive
 - Urine urobilinogen absent
- ALT and AST – 100/80 ↑ (Associated hepatocellular damage in long standing cholestasis)
- **ALP and GGT- 540/300 ↑ ↑ ↑ ↑** (Extra-hepatic cholestasis)
- Serum albumin – Normal (Albumin synthesis not affected)
- Prothrombin time – Increased (decreased vit K absorption, due to cholestasis)
- Obstructive/ Cholestatic Jaundice (Extrahepatic cholestasis/ Posthepatic)
 - Gall stone in bile duct, cancer head of pancreas (Extrahepatic cholestasis)

Case report 3

A 45-year-old man brought in comatose state. 1-day history of hematemesis (vomiting blood). Repeated abdominal distension and ascitic fluid removal for past 3 years; Alcohol use for 20-25 years

Laboratory parameter	Patient values	Reference range
Serum total bilirubin	3.6 mg/dL	0.1-1.0 mg/dL
Serum conjugated (direct) bilirubin	1.5 mg/dL	0.0-0.3 mg/dL
ALT	120	10-40 U/L
AST	300	12-38 U/L
ALP	180	25-100 U/L
GGT	190	5-40 U/L
Serum albumin	2.1 g/dL	3.5-5.5 g/dL
Prothrombin time	35 seconds	11-15 seconds
Ammonia	300 µg/dL	15-45 µg/dL

Case report 3

- A 45-year-old man brought in comatose state. 1-day history of hematemesis (vomiting blood)
- Repeated abdominal distension and ascitic fluid removal for past 3 years
- Alcohol abuse for 20-25 years
- Laboratory tests (**Chronic alcoholic cirrhosis**)
 - Serum bilirubin 3.6mg/dL
 - Serum conjugated bilirubin 1.5mg/dL (Mixed bilirubin)
 - Serum albumin 2.1g/dL (indicating long-standing liver disease)
 - Serum globulins were elevated
 - Serum AST (300)>>ALT (120)[ratio AST:ALT > 2] (indicating chronic alcoholic cirrhosis)
 - ALP - 180↑ (indicating intrahepatic cholestasis)
 - GGT- 190 ↑ ↑ (enzyme induction by alcohol)
 - Prothrombin time - ↑ (decreased capacity of liver to synthesize clotting factors)
 - Serum ammonia – Elevated (reduced conversion of ammonia to urea) → **Encephalopathy**

Summary of laboratory findings in common liver diseases

Hepatic jaundice

Answers in Practice Questions

Laboratory findings	Acute hepatocellular damage (Case 1)	Chronic alcoholic cirrhosis (Case 3)	Obstructive/ Posthepatic/ Cholestatic jaundice (Case 2)
Serum total bilirubin			
Serum conjugated bilirubin			
Serum unconjugated bilirubin			
ALT			
AST			
ALP			
GGT			
Serum albumin			